Unit 3: Practical Skills in Biology I

IAS compulsory unit

Externally assessed

Unit description

Introduction	Students are expected to develop experimental skills, and a knowledge and understanding of experimental techniques, by carrying out the core practicals and other recommended practical investigations and experiments while they study units 1 and 2. This will require them to work safely, produce valid results and present data in the most appropriate format. This unit will assess students' ability to apply their knowledge and understanding of experimental design, procedures and techniques developed throughout units 1 and 2.
Practical skills identified for assessment	• Solve problems set in practical contexts. • Apply scientific knowledge to practical contexts. • Comment on experimental design and evaluate scientific methods. • Present data in appropriate ways. • Evaluate results and draw conclusions with reference to measurement uncertainties and errors. • Identify variables, including those that must be controlled. • Plot and interpret graphs. • Process and analyse data using appropriate mathematical skills (see *Appendix 6: Mathematical skills and exemplifications*). • Know and understand how to use a wide range of apparatus, materials and techniques safely, appropriate to the knowledge and understanding in this specification. • Plan an investigation to test a hypothesis.
Practical skills to be developed through teaching and learning	• Apply investigative approaches and methods to practical work. • Use a range of practical equipment and materials safely and correctly. • Follow written instructions. • Make and record observations. • Present information and data in a scientific way. • Use appropriate software and tools to collect and process data. • Use online and offline research skills, including websites, textbooks and other printed scientific sources of information. • Cite sources of information correctly. • Use a wide range of experimental and practical instruments, equipment and techniques appropriate to the knowledge and understanding included in this specification.

Assessment information

- First assessment: June 2019.
- The assessment is 1 hour and 20 minutes.
- The assessment is out of 50 marks.
- Candidates must answer all questions.
- The paper may include calculations, short open, open-response and calculation questions.
- The paper will include a minimum of 5 marks that target mathematics at Level 2 or above.
- Calculators may be used in the examination. Please see *Appendix 8: Use of calculators*.

Planning

Questions on Unit 3 may require students to plan and evaluate an investigation, based on the practical activities included in units 1 and 2.

Candidates will be assessed on their ability to:

Plan an experiment

- identify the apparatus required
- identify the dependent and independent variables, standardised or controlled variables
- describe how to measure relevant variables using the most appropriate instrument and correct measuring techniques
- identify and state how to control all other relevant variables to make it a fair test
- discuss whether repeat readings are appropriate
- identify health and safety issues and discuss how they may be dealt with
- discuss how the data collected will be used
- identify possible sources of uncertainty and/or systematic error, and explain how they may be reduced or eliminated
- comment on the implications of biology (for example: benefits/risks) and on its context (for example: social/environmental/historical).

Implementation and measurements

Candidates will be assessed on their ability to:

Implementation and measurements	• comment on the number of readings taken • comment on the range of measurements taken • comment on significant figures • check a reading that is inconsistent with other readings – for example, a point that is not on the line of a graph • comment on how the experiment may be improved, possibly by using additional apparatus (for example, to reduce errors).

Processing results

Candidates may be provided with experimental data in a tabulated or graphical form.

Candidates will be assessed on their ability to:

Processing results	• perform calculations, using the correct number of significant figures • plot results on a graph using an appropriate scale • use the correct units throughout • comment on the relationship obtained from the graph • determine the relationship between two variables or determine a constant with the aid of a graph – for example, by determining the gradient using a large triangle • suggest realistic modifications to reduce errors • suggest realistic modifications to improve the experiment • discuss uncertainties, qualitatively and quantitatively.

Unit 4: Energy, Environment, Microbiology and Immunity

IA2 compulsory unit

Externally assessed

Unit description

Introduction	This unit begins with energy capture in photosynthesis and the synthesis of organic compounds by plants, and the flow of energy in ecosystems. This is followed by a consideration of the carbon cycle and how disruption of this cycle may lead to climate change. Students will also consider changes that occur in populations, both in the short term and long term, as a result of mutation and natural selection. The unit continues with an introduction to the diversity and features of microorganisms and how hosts respond to infection by pathogens. This leads to a consideration of the role of microorganisms in decomposition of organic materials and the techniques and applications of polymerase chain reaction (PCR) and gel electrophoresis.
Practical skills	In order to develop their practical skills, students should be encouraged to carry out a range of core practical experiments related to this topic.
Mathematical skills	There are opportunities for the development of mathematical skills in this unit, including tabulation and graphical treatment of data, understanding the principles of sampling, exponential and logarithmic functions and the use of statistics. (Please see *Appendix 6: Mathematical skills and exemplifications* for further information.)

Assessment information

- First assessment: January 2020.
- The assessment is 1 hour and 45 minutes.
- The assessment is out of 90 marks.
- Candidates must answer all questions.
- The paper may include multiple-choice, short-open, open-response, calculations and extended-writing questions.
- The paper will include synoptic questions that may draw on two or more different topics.
- The paper will include a minimum of 9 marks that target mathematics at Level 2 or above.
- Calculators may be used in the examination. Please see *Appendix 8: Use of calculators*.
- Candidates will be expected to apply their knowledge and understanding to familiar and unfamiliar contexts.